

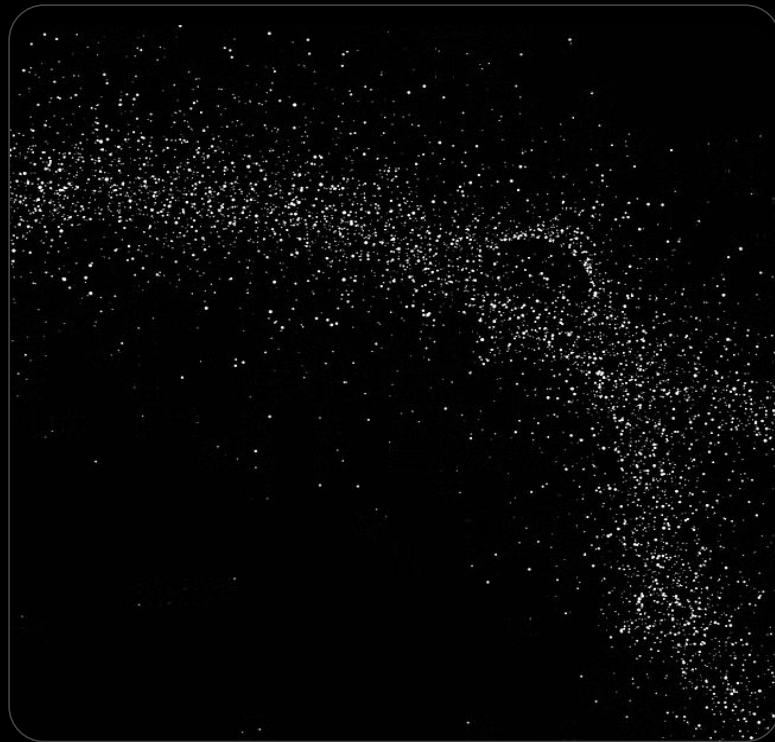
AI + Misinformatio n

~~Modeling~~ Theme

August 2026

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Project Overview

Our Problem

Artificial Intelligence is becoming increasingly common in enterprise applications such as healthcare, finance, customer service, and education. Although these systems can generate helpful responses, they may also produce hallucinations, responses that sound convincing but contain unsupported or incorrect information.

These hallucinations can reduce trust in AI and may lead to poor decision making in high-impact environments.

Project Goals

01. Detect Hallucinations

Identify and flag AI-generated responses using advanced machine learning models.

02. Model Comparison

Evaluate and compare different classification algorithms to find the most accurate detector.

03. Streamlit Deployment

Deploy the best-performing model into a functional, user-friendly web application.

Why this Project Matters?

Why Hallucination Detection Is Important?

- AI-generated misinformation can influence important decisions.
- Organizations need reliable AI systems they can trust.
- Detecting hallucinations improves AI safety and transparency.
- A reliable detector can reduce misinformation in finance, healthcare, education, and customer support.

REAL-WORLD APPLICATIONS

01 Healthcare

02 Banking

03 Customer Support

04 Enterprise AI

05 Education

Proposed Research Question

The Inquiry

"Can a single hallucination detector be deployed across different enterprise applications, or does accuracy degrade significantly by domain?"

Project Objectives

01. Train ML Models

Train machine learning models on the RAGTruth dataset.

02. Compare Algorithms

Compare multiple classification algorithms to evaluate baseline accuracy.

03. Performance Metrics

Evaluate model performance using several comprehensive metrics.

04. Streamlit Deployment

Deploy the best-performing model for real-time user interaction.

How Our Project Evolved

01 Initial Proposal

Focused on identifying linguistic features that distinguish hallucinated AI-generated responses.

02 Mid-Project

- Explored the RAGTruth dataset using EDA.
- Performed feature engineering with TF-IDF and text-based features.
- Trained Logistic Regression and Decision Tree models.

03 Final Outcome

- Logistic Regression achieved the best performance (~74.5% accuracy).
- Successfully deployed the model using Streamlit.
- Built an interactive application for hallucination detection.

Dataset & Data Exploration

01 Dataset Overview

Source:

RAGTruth - ParticleMedia / ACL 2024
(NewsBreak + UIUC)

Size:

17,790 responses with 9 original features, plus 1 binary target (has_hallucination)

Labels:

Hallucinated, Factual

Train/Test Split:

80% Training, 20% Testing

02 Data Exploration

Conducting rigorous EDA to uncover core patterns:

- Distribution of hallucinated vs factual responses
- Analysis of response length
- Domain distribution across sources

03 Feature Engineering

Extracting predictive signals for model training:

- TF-IDF text features
- Contextual similarity score
- Response length metrics
- Named Entity Recognition (NER)

Exploratory Data Analysis (EDA)



In looking at the descriptive statistics done in regards to the research question, the following data was divided by powerful LLMs on the market:

Hallucination Response

- Chat-4.0 had over 401 Responses
- Llama-2-7B Chat had over 1,832
- Mistral-7B-Instruct had 1,953

Hallucination Spans

- Chat-4.0 had 533 Responses
- Llama-2-7B had 3,302
- Mistral-7B-Instruct had 3,562

Hallucination Response (response-level):

This is a binary, whole-answer judgement: each response received an overall label indicating whether it contains hallucinated information, where annotators marked the specific spans of generated text containing hallucinations

Hallucination Span (Span/word-level):

This is a fine-grained unit: the exact stretch of text inside a response that conflicts with or is unsupported by the retrieved context. Annotators marked each hallucinated span and categorized it into one of four types: conflict, subtle conflict, evident introduction of baseline information, and subtle introduction of baseline information

Machine Learning Algorithm(s) Strategy

01 Logistic Regression

Type: Supervised Learning | Binary Classification

Input: TF-IDF Features

Output: Hallucinated / Factual

Pros

Fast, highly interpretable, and delivers strong baseline accuracy.

Cons

Limitation: Assumes linear relationships between features.

02 Decision Tree

Type: Supervised Learning | Binary Classification

Pros

Very easy to interpret, map visually, and explain to stakeholders.

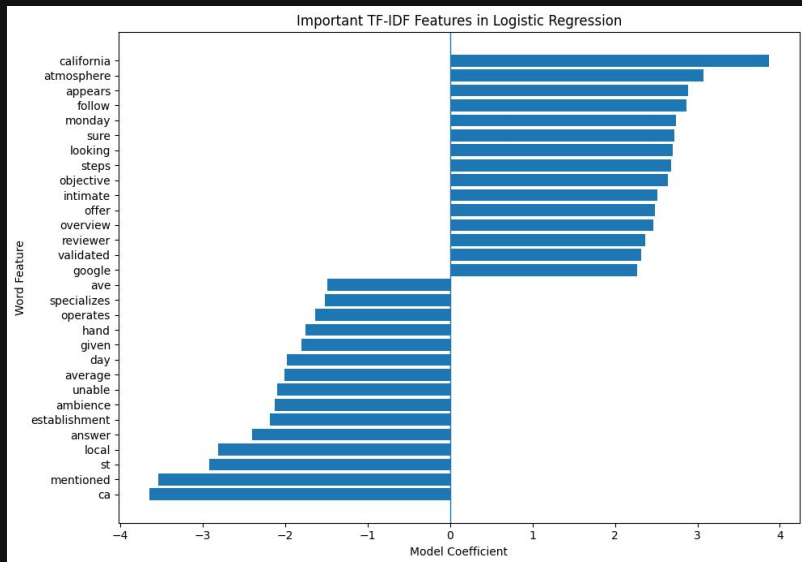
Cons

High risk of overfitting, especially with high-dimensional data.

Feature Importance

These visualizations explain which TF-IDF word features had the greatest influence on each model's predictions

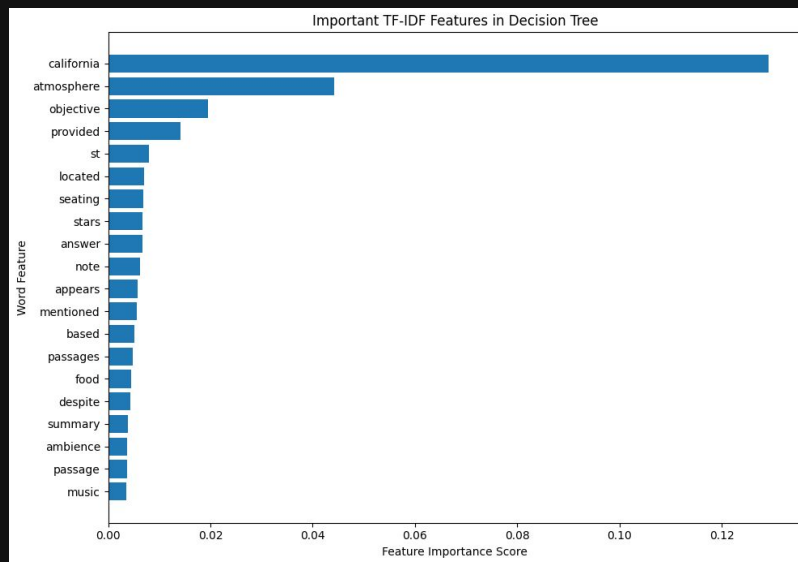
Logistic Regression Feature Importance



Methodology: Uses coefficient weights to identify influential TF-IDF features.

Interpretation: **Positive values** indicate hallucinated features, while **negative values** point to factual ones.

Decision Tree Feature Importance



Methodology: Uses feature importance scores to rank influential TF-IDF features.

Interpretation: Higher scores indicate a greater, more direct impact on model decisions.

Summary: 2. How Will Data Be Used? 3. Potential Impact

02 How Will Data be Used?

We will use the RAGTruth dataset which contains AI-generated responses labeled as hallucinated or factual. The dataset will be analyzed to identify linguistic features that distinguish hallucinated responses from supported claims. We will also evaluate whether models trained on one domain perform well across different domains.

03 Potential Impact

The way we anticipate our findings to help people make better decisions is by educating them on the topic of AI hallucination. Once they are educated, our findings will also explain the different ways AI hallucination could be shown or at least manifested, especially in specific enterprise domain media, such as finance, healthcare, etc.

Number of Hallucinated Responses by Model

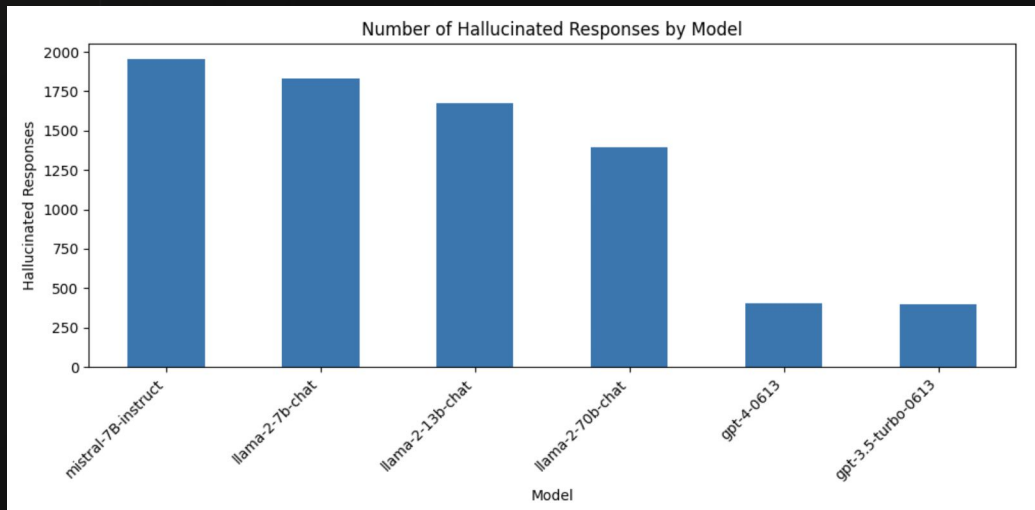
KEY OBSERVATION

GPT-4 Generated the Fewest Hallucinations

GPT-4 produced the fewest hallucinated responses, demonstrating superior performance in maintaining factual consistency.

Conversely, Mistral-7B-Instruct and LLaMA-2 models produced a significantly higher volume of hallucinations across the test sets.

HALLUCINATION COUNT BY LLM



Performance Metrics: Logistic Regression vs. Decision Tree

01. Logistic Regression (Best Performer)

Logistic Regression Results

Accuracy: 0.7453625632377741

	precision	recall	f1-score	support
False	0.72	0.90	0.80	2025
True	0.80	0.55	0.65	1533
accuracy			0.75	3558
macro avg	0.76	0.72	0.72	3558
weighted avg	0.76	0.75	0.74	3558
[[1815 210] [696 837]]				

02. Decision Tree Results

Decision Tree Results

Accuracy: 0.6793142214727375

	precision	recall	f1-score	support
False	0.71	0.73	0.72	2025
True	0.63	0.61	0.62	1533
accuracy			0.68	3558
macro avg	0.67	0.67	0.67	3558
weighted avg	0.68	0.68	0.68	3558
[[1477 548] [593 940]]				

METRIC	LOGISTIC REG.	DECISION TREE
Accuracy	74.5%	67.0%
Precision	0.76	0.67
Recall	0.72	0.67
F1 Score	0.72	0.67

Deployment

Interactive Tool

This deployment demonstrates how our machine learning model operates in production.

Users can input AI-generated text and instantly evaluate its faithfulness using our trained Logistic Regression classifier.

Our deployed application allows users to evaluate AI-generated responses in real time using the trained Logistic Regression model.

01. INPUT INTERFACE

AI Hallucination Detector

Paste an AI-generated response below. Our Logistic Regression model will estimate whether it is faithful or potentially hallucinated.

AI-generated response

Paste the response here...

Analyze response

About the model

The application uses a Logistic Regression classifier trained on TF-IDF text features from the RAGTruth dataset. During evaluation, the model achieved approximately 74% accuracy.

02. LIVE EVALUATION

AI Hallucination Detector

Paste an AI-generated response below. Our Logistic Regression model will estimate whether it is faithful or potentially hallucinated.

AI-generated response

The Earth revolves around the Sun once every 365 days

Analyze response

Prediction: Faithful

Model confidence

82.5%

This prediction is produced by a machine-learning model and should not replace fact-checking or human review.

About the model

The application uses a Logistic Regression classifier trained on TF-IDF text features from the RAGTruth dataset. During evaluation, the model achieved approximately 74% accuracy.

Challenges, Lessons Learned & Future Work

Challenges

- Preparing text data for machine learning.
- Connecting the trained model to Streamlit.
- Deploying through GitHub.

Lessons Learned

- Feature engineering significantly improves text classification.
- Logistic Regression outperformed Decision Tree.
- Deployment is an important part of an ML project.

Future Work

- Train using additional enterprise datasets.
- Test Random Forest and newer classifiers.
- Improve feature engineering.
- Increase model accuracy.
- Add explainable AI features.

How the results connect to the overall AI4ALL Essential Question?

"How can the development process of responsible AI/ML solutions address and mitigate the technology's negative impacts while strengthening its positive effects?"

Based on our data collected and analyses, the best way to answer this question is that most treatments of "responsible AI" fail because they operate at the wrong level of abstraction - principles like fairness, transparency, and accountability are stated as values rather than as decisions someone makes. Hallucination is valuable precisely because it collapses that gap. A hallucination is simultaneously a technical defect, an epistemic harm to a user, and a quantifiable event.

Conclusion

Key Takeaways

- Successfully built and evaluated a hallucination detection model.
- Logistic Regression achieved the best performance (~74.5% accuracy).
- Deployed an interactive Streamlit application.
- Demonstrated how responsible AI can improve reliability and reduce misinformation.

Thank You!

Questions?

Citations

1. Niu, C., Wu, Y., Zhu, J., Xu, S., Shum, K., Zhong, R., ... Zhang, T. (n.d.). Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. Retrieved from <https://aclanthology.org/2024.acl-long.585/>
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